

Problem Statement

Write a C program to demonstrate the order of execution of operators in an expression using operator precedence and associativity.

Program

```
#include <stdio.h>
```

```
int main()
{
    int a = 10, b = 5, c = 2;
    int result;

    result = a + b * c;

    printf("Expression: a + b * c\n");
    printf("a = %d, b = %d, c = %d\n", a, b, c);
    printf("Result = %d\n", result);

    return 0;
}
```

Output

Expression: a + b * c

a = 10, b = 5, c = 2

Result = 20

Explanation: Multiplication * has higher precedence than addition +, so b * c is evaluated first.

2. Program to Check Whether a Number is Even or Odd

Problem Statement

Write a C program to accept a number from the user and check whether the given number is even or odd.

Program

```
#include <stdio.h>

int main()
{
    int num;

    printf("Enter a number: ");
    scanf("%d", &num);

    if (num % 2 == 0)
    {
        printf("%d is an Even number.\n", num);
    }
    else
    {
        printf("%d is an Odd number.\n", num);
    }

    return 0;
}
```

Output

```
Enter a number: 25
25 is an Odd number.
```

3. Program to Determine Whether a Number is Positive, Negative, or Zero

Problem Statement

Write a C program to accept a number from the user and determine whether it is positive, negative, or zero.

Program

```
#include <stdio.h>

int main()
{
    int num;

    printf("Enter a number: ");
    scanf("%d", &num);

    if (num > 0)
    {
        printf("%d is a Positive number.\n", num);
    }
}
```

```

    }
    else if (num < 0)
    {
        printf("%d is a Negative number.\n", num);
    }
    else
    {
        printf("The number is Zero.\n");
    }

    return 0;
}

```

Output

Enter a number: -15

-15 is a Negative number.

4. Menu-Driven Calculator Using Switch-Case

Problem Statement

Write a C program using the switch-case statement to create a menu-driven calculator that repeatedly performs arithmetic operations until the user chooses to exit.

Program

```
#include <stdio.h>
```

```

int main()
{
    int choice;
    float num1, num2, result;

    do
    {
        printf("\n----- MENU-DRIVEN CALCULATOR ----- \n");
        printf("1. Addition\n");
        printf("2. Subtraction\n");
        printf("3. Multiplication\n");
        printf("4. Division\n");
        printf("5. Exit\n");

        printf("Enter your choice: ");
        scanf("%d", &choice);

        if (choice >= 1 && choice <= 4)
        {
            printf("Enter two numbers: ");
            scanf("%f %f", &num1, &num2);
        }

        switch (choice)

```

```
{
    case 1:
    {
        result = num1 + num2;
        printf("Result = %.2f\n", result);
        break;
    }

    case 2:
    {
        result = num1 - num2;
        printf("Result = %.2f\n", result);
        break;
    }

    case 3:
    {
        result = num1 * num2;
        printf("Result = %.2f\n", result);
        break;
    }

    case 4:
    {
        if (num2 != 0)
        {
            result = num1 / num2;
            printf("Result = %.2f\n", result);
        }
        else
        {
            printf("Division by zero is not possible.\n");
        }
        break;
    }

    case 5:
    {
        printf("Exiting the calculator...\n");
        break;
    }

    default:
    {
        printf("Invalid choice. Please try again.\n");
    }
}
```

```
    } while (choice != 5);  
  
    return 0;  
}
```

Output

----- MENU-DRIVEN CALCULATOR -----

1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit

Enter your choice: 1

Enter two numbers: 20 10

Result = 30.00

----- MENU-DRIVEN CALCULATOR -----

1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit

Enter your choice: 3

Enter two numbers: 5 4

Result = 20.00

----- MENU-DRIVEN CALCULATOR -----

1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit

Enter your choice: 5

Exiting the calculator...

Switch case calculator program:

#include <stdio.h>

```
int main()  
{  
    int choice;  
    float a, b;  
  
    printf("1. Addition\n");  
    printf("2. Subtraction\n");  
    printf("3. Multiplication\n");  
    printf("4. Division\n");  
  
    printf("Enter your choice: ");
```

```
scanf("%d", &choice);

printf("Enter two numbers: ");
scanf("%f %f", &a, &b);

switch(choice)
{
    case 1:
        printf("Result = %.2f", a + b);
        break;

    case 2:
        printf("Result = %.2f", a - b);
        break;

    case 3:
        printf("Result = %.2f", a * b);
        break;

    case 4:
        printf("Result = %.2f", a / b);
        break;

    default:
        printf("Invalid choice");
}

return 0;
}
```

5. Multiplication Table Using While, Do-While, and For Loops

Problem Statement

Write a C program to generate the multiplication table of any number using while loop, do-while loop, and for loop.

Program

```
#include <stdio.h>
```

```
int main()
{
    int num;
    int i;
```

```

printf("Enter a number: ");
scanf("%d", &num);

printf("\nMultiplication Table Using While Loop:\n");

i = 1;

while (i <= 10)
{
    printf("%d x %d = %d\n", num, i, num * i);
    i++;
}

printf("\nMultiplication Table Using Do-While Loop:\n");

i = 1;

do
{
    printf("%d x %d = %d\n", num, i, num * i);
    i++;
} while (i <= 10);

printf("\nMultiplication Table Using For Loop:\n");

for (i = 1; i <= 10; i++)
{
    printf("%d x %d = %d\n", num, i, num * i);
}

return 0;
}

```

Output

Enter a number: 5

Multiplication Table Using While Loop:

```

5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50

```

Multiplication Table Using Do-While Loop:

$5 \times 1 = 5$
 $5 \times 2 = 10$
 $5 \times 3 = 15$
 $5 \times 4 = 20$
 $5 \times 5 = 25$
 $5 \times 6 = 30$
 $5 \times 7 = 35$
 $5 \times 8 = 40$
 $5 \times 9 = 45$
 $5 \times 10 = 50$

Multiplication Table Using For Loop:

$5 \times 1 = 5$
 $5 \times 2 = 10$
 $5 \times 3 = 15$
 $5 \times 4 = 20$
 $5 \times 5 = 25$
 $5 \times 6 = 30$
 $5 \times 7 = 35$
 $5 \times 8 = 40$
 $5 \times 9 = 45$
 $5 \times 10 = 50$